

STA 445 S26 Assignment 5

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Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010. Sept is not recognized.

```
mdy("Sept 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs. The month needs more than one letter to specify what month it is.

```
mdy("S 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

e. 07-Dec-1941.

```
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong. 1 and 5 can either be the month or day.

```
mdy("1-5-1998")
```

```
## [1] "1998-01-05"
```

g. 21-5-1998. Comment on why you know you are correct. 21 is too high to represent the 12 months in a year we have, so it should be the day not month, while 5 can represent the month.

```
dmy("21-5-1998")
```

```
## [1] "1998-05-21"
```

h. 2020-May-5 10:30 am

```
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```
ymd_hm("2020-May-5 10:30 am", tz="US/Pacific" )
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```
ymd_hm("2020-May-5 10:30 am", tz="America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```
myBirthday <- mdy("Feb 21, 2006")
todayDate <- mdy("Mar 17, 2026")
```

```
myBirthday + years(64)
```

```
## [1] "2070-02-21"
```

b. Calculate your current age (in years).

```
currentAge <- year(as.period(myBirthday %--% todayDate))
currentAge
```

```
## [1] 20
```

c. Using your result in part (b), calculate the date of your next birthday.

```
nextBirthday <- myBirthday + years(currentAge + 1)
nextBirthday
```

```
## [1] "2027-02-21"
```

d. The number of *days* until your next birthday. 341 days till my next birthday

```
nextBirthday - todayDate
```

```
## Time difference of 341 days
```

e. The number of *months* and *days* until your next birthday. number of months is 11 and the number of days are 4.

```
(as.period(todayDate %--% nextBirthday, unit = "months"))
```

```
## [1] "11m 4d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there? It will be 10 AM on May 9, 2015.

```
mdy_h("May 8, 2015 3 PM", tz = "US/Arizona") %>%
  with_tz(tz="Pacific/Auckland")
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

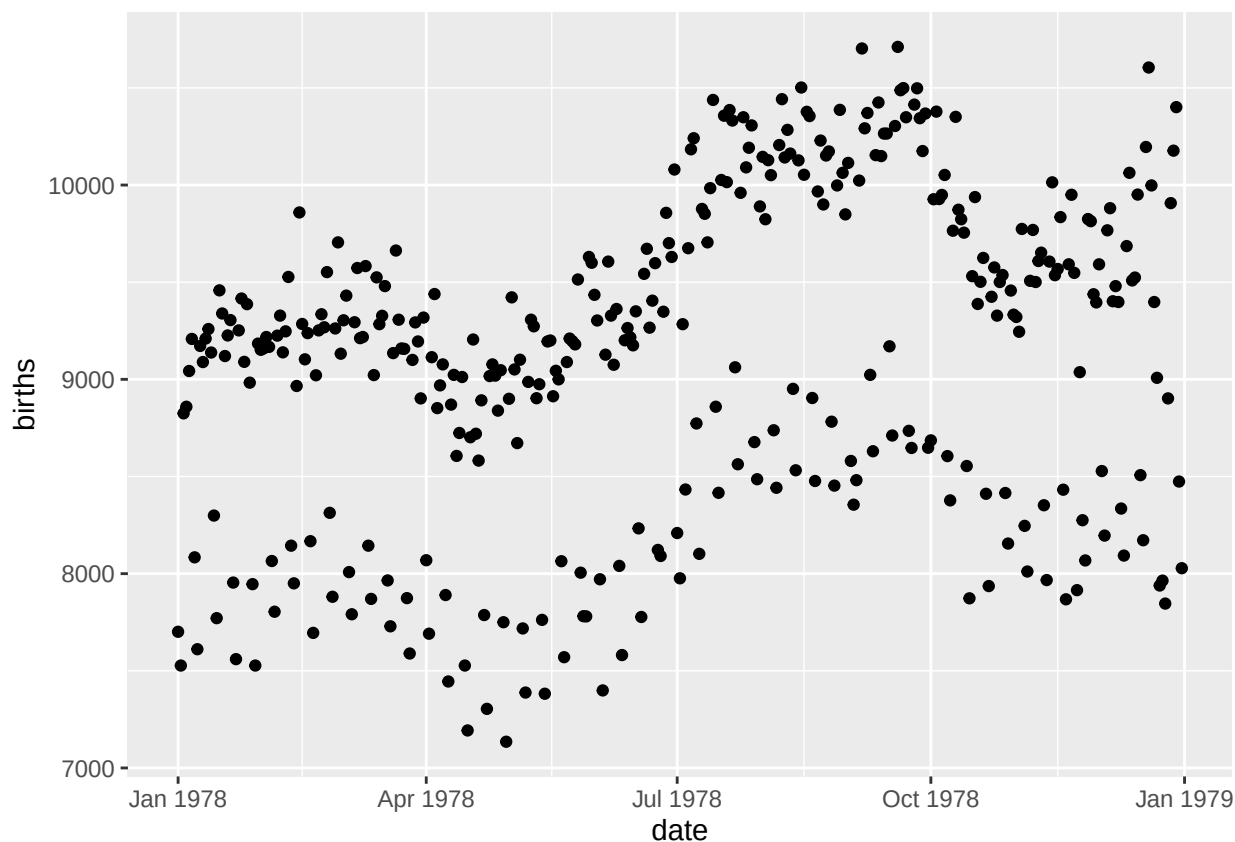
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

- Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```
data('Births78', package='mosaicData')
myBirths78 <- Births78 %>% select(date, births)
```

- Graph the number of `births` vs the `date` with `date` on the x-axis. What stands out to you? Why do you think we have this trend? There is a up and down, or zig-zag pattern in the graph. There is the bottom layer that has less data points, but also the same pattern as the top layer, which has more data points. I think we have this trend due to the time of when a birth happens. Lets say the top is on weekdays because more people means more data, and the bottom layer is on weekends because less people give birth on those days.

```
ggplot(data=myBirths78, aes(x=date, y=births)) +
  geom_point()
```



- To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
myBirths78 <- myBirths78 %>%
  mutate(dow = wday(date, label=TRUE, abbr=FALSE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(data=myBirths78, aes(x=date, y=births, color=dow)) +
  geom_point()
```

